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AI for Bharat Hackathon

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Problem Statement :India is experiencing an unprecedented surge in cyber frauds, with scammers targeting vulnerable populations through sophisticated social engineering attacks. Digital arrest scams, fake police calls, bank impersonation, and other fraudulent schemes are causing significant financial and emotional harm to citizens. Currently, Citizens lack Real-Time, accessible mechanisms to access the legitimacy of incoming calls before falling victim to such frauds.

Brief about the Idea:

The proposed solution is an AI-powered fraud risk advisory system that protects citizens from phone-based cyber crimes such as digital arrest scams and financial impersonation frauds. It analyzes historical call behavior, reported fraud data, and calling patterns to assess the risk of incoming calls in real time. When a call is potentially suspicious, the system provides a clear advisory warning to help users make informed decisions before engaging or transferring money. Designed as a non-blocking, ethical solution, it prioritizes privacy, user trust, and explainable alerts. The solution focuses on public impact by safeguarding common citizens, senior citizens, and digitally vulnerable communities across Bharat.

Why AI is required in this solution?

AI is required because phone fraud is not limited to fixed keywords or blacklisted numbers. Scammers continuously change their language, tone, pressure tactics, and impersonation patterns. An AI layer helps the system understand the meaning and intent behind a call transcript, such as urgency, social engineering, bank impersonation, OTP requests, digital arrest threats, and payment coercion. This makes the solution more adaptive, scalable, and effective against new fraud variants that are not already hardcoded into the system.

How AWS services are used within our architecture?

The solution is built as a cloud-native fraud detection pipeline on AWS. Amazon API Gateway exposes the fraud-analysis endpoint and receives incoming requests from the client application. AWS Lambda contains the fraud analysis engine, which processes the phone number and transcript, computes the fraud score, and generates the risk classification. Amazon DynamoDB is used for persistent storage in two layers: `fraud_logs` stores each analyzed call event, while `phone_reputation` stores scam reputation information for repeat caller detection.

What value the AI layer adds to the user experience?

The AI layer improves user experience by providing smarter and more trustworthy scam warnings instead of basic spam alerts. It helps the user understand why a call is risky by identifying fraud indicators such as OTP requests, bank impersonation, urgency, threats, and payment demands. This allows the system to give more meaningful, context-aware alerts and recommendations, which increases user confidence and reduces the chance of falling for new scam patterns. In the future, this AI layer can also support real-time call assistance, multilingual fraud detection, and personalized scam-risk explanations for users from Bharat.

List of features offered by the solution

- **AI-based fraud risk scoring** – Analyzes call transcripts to detect potential scams and assigns a fraud risk score.
- **Fraud pattern detection** – Identifies common scam indicators such as OTP requests, bank impersonation, urgency, payment requests, and threat language.
- **Real-time call risk analysis** – Allows a call transcript or call event to be analyzed instantly through an API.
- **Phone number reputation tracking** – Maintains a reputation score for each phone number to identify repeat scam callers.
- **Crowdsourced fraud intelligence** – Multiple reports related to the same number increase its fraud reputation score, enabling detection of emerging scam numbers.
- **Fraud activity logging** – Stores analyzed calls and risk indicators in a database for monitoring and analytics.
- **Scalable serverless architecture** – Built using AWS services such as API Gateway, Lambda, and DynamoDB for reliable and scalable operation.
- **Actionable risk alerts** – Classifies calls into LOW, MEDIUM, or HIGH risk categories to help users make informed decisions.
- **Extensible AI layer** – The architecture allows future integration of advanced AI models for NLP-based scam detection and multilingual fraud analysis.

Process Flow Diagram

```
User / Mobile App
        |
        | 1. Send call transcript + phone number
        ▼
API Gateway (REST API)
        |
        | 2. Receives POST /analyze-call request
        ▼
AWS Lambda - Fraud Analyzer
        |
        | 3. Process transcript
        | 4. Detect fraud indicators
        |     • OTP request
        |     • Bank impersonation
        |     • Urgent language
        |     • Payment request
        |     • Threat language
        |
        | 5. Calculate fraud risk score
        ▼
Fraud Scoring Engine
        |
        | 6. Determine risk level
        |    LOW / MEDIUM / HIGH
        ▼
```

```
DynamoDB Database
        |
        ├── Store call record → fraud_logs
        |
        └── Update phone reputation → phone_reputation
                |
                └── Detect repeat scam callers
        ▼
API Response
        |
        | 7. Return fraud score + risk level
        ▼
User / Application
        |
        └── Display fraud alert to user
```

Wireframes/Mock diagrams of the proposed solution

1. Home Screen – Fraud Protection Dashboard

```
+-----------------------------------+
|      Fraud Shield Protection      |
|-----------------------------------|
| Protection Status: ACTIVE         |
|                                   |
| Calls Analyzed Today: 12          |
| Fraud Alerts Detected: 4          |
|                                   |
|  [ Check Call ]                   |
|  [ Report Scam Number ]           |
|                                   |
| Recent Alerts                     |
| - High risk call from 9876543210 |
| - Suspicious call from 9123456789|
+-----------------------------------+
```

2. Incoming Call Analysis Screen

```
+-----------------------------------+
| Incoming Call Detected            |
|-----------------------------------|
| Caller Number: +91 9876543210     |
|                                   |
| Analyzing call transcript...      |
|                                   |
| Fraud Indicators Detected:        |
| • OTP request                     |
| • Bank impersonation              |
| • Urgent language                 |
|                                   |
| Risk Score: 85 / 100              |
| Risk Level: HIGH                  |
|                                   |
| [ Block Number ]  [ Report ]      |
+-----------------------------------+
```

3. Scam Number Reputation Screen

```
+-----------------------------------+
| Phone Reputation Check            |
|-----------------------------------|
| Number: +91 9876543210            |
|                                   |
| Fraud Score: 88                   |
| Number of Reports: 12             |
| Risk Level: HIGH                  |
|                                   |
| Recent Fraud Types                |
| • OTP Scam                        |
| • Bank Impersonation              |
|                                   |
| [ Block Number ]                  |
+-----------------------------------+
```

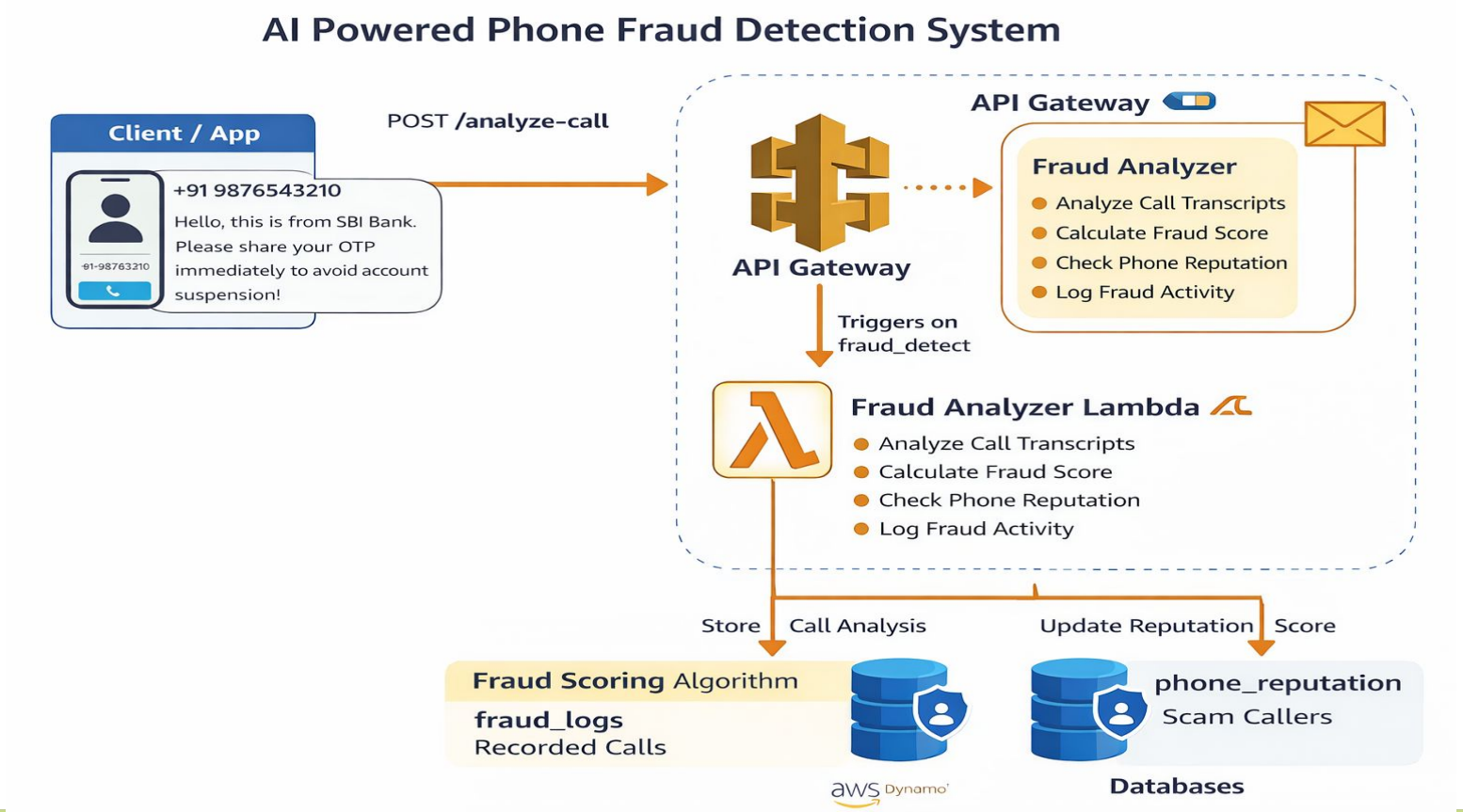
4. Report Fraud Screen

```
+-----------------------------------+
| Report Scam Call                  |
|-----------------------------------|
| Phone Number: [__________]        |
|                                   |
| Scam Type:                        |
|  o OTP Scam                       |
|  o Bank Fraud                     |
|  o Digital Arrest Scam            |
|  o UPI Fraud                      |
|                                   |
| Description:                      |
| [_________________________]       |
|                                   |
| [ Submit Report ]                 |
+-----------------------------------+
```

5. Fraud Analytics Dashboard

```
+-----------------------------------------+
| Fraud Detection Analytics               |
|-----------------------------------------|
| Total Calls Analyzed: 2,450             |
| Fraud Alerts Generated: 320             |
| High Risk Numbers Detected: 65          |
|                                         |
| Top Scam Categories                     |
| - OTP Fraud                             |
| - Bank Impersonation                    |
| - Digital Arrest Scam                   |
|                                         |
| Fraud Network Graph                     |
| [Visualization of connected scam numbers] |
+-----------------------------------------+
```

Architecture diagram of the proposed solution:



Technologies utilized in the solution:

1.**AWS Lambda** – Executes the fraud detection logic and processes incoming requests in a serverless environment.

2.**Amazon API Gateway** – Provides REST API endpoints for receiving call transcript data and forwarding requests to the backend.

3.**Amazon DynamoDB** – Stores fraud detection logs and phone-number reputation data for repeat scam detection.

4.**Node.js** – Used for implementing the fraud analysis engine and API processing logic.

5.**AWS SDK (JavaScript)** – Enables Lambda functions to interact with DynamoDB and other AWS services.

Fraud Detection Logic

6.**Rule-based Fraud Scoring Algorithm** – Detects scam indicators such as OTP requests, bank impersonation, urgent language, and payment demands.

7.**Phone Reputation Tracking** – Maintains a dynamic risk score for phone numbers to identify repeat scam callers.

8..**REST API Architecture** – Enables scalable communication between the client application and backend services.

9.**Postman** – Used for testing and validating API endpoints during development.

Snapshots of the prototype:

Code source Info

Open in Visual Studio Code | Upload from

fraud-analyzer

EXPLORER

DEPLOY ✓ Current

Deploy (Ctrl+Shift+U)

Test (Ctrl+Shift+I)

TEST EVENTS [SELECTED: T...

+ Create new test...

Private saved ev...

PROBLEMS | OUTPUT | CODE REFERENCE LOG | TERMINAL

Execution Results

```
Status: Succeeded
Test Event Name: test11

Response:
{
  "statusCode": 200,
  "headers": {
    "Content-Type": "application/json",
    "Access-Control-Allow-Origin": "*"
  },
  "body": "{\"phone_number\":\"9876543219\",\"fraud_score\":115,\"risk_level\":\"HIGH\",\"indicators_detected\":
  [\"otp_request\",\"bank_impersonation\",\"urgency\",\"money_request\",\"threat_language\"],\"number_reputation\":
  {\"fraud_score\":115,\"number_of_reports\":1,\"risk_level\":\"HIGH\"}}"
}
```

Table: fraud_logs - Items returned (6)

Actions | Create item

Scan started on March 08, 2026, 21:29:19

1

call_id *(String)*	fraud_score	indicators_detected
1772969109290	115	[{ "S" : "otp_request" }, ...
1772965979607		
1772985435209	115	[{ "S" : "otp_request" }, ...
1772970582191	115	[{ "S" : "otp_request" }, ...
1772969570326	115	[{ "S" : "otp_request" }, ...
1772912654693		

```
Status: Succeeded
Test Event Name: TEST121

Response:
{
  "statusCode": 200,
  "headers": {
    "Content-Type": "application/json",
    "Access-Control-Allow-Origin": "*"
  },
  "body": "{\"phone_number\":\"9023134211\",\"fraud_score\":40,\"risk_level\":\"MEDIUM\",\"indicators_detected\":
  [\"money_request\",\"threat_language\"],\"number_reputation\":{\"fraud_score\":40,\"number_of_reports\":1,
  \"risk_level\":\"MEDIUM\"}}"
}

The area below shows the last 4 KB of the execution log.
```

1

phone_number *(String)*	fraud_score	last_seen	numl
9876543219	78	177298573...	2
9876543210	115	177297058...	2
9023134211	40	177298586...	1

Table: phone_reputation - Items returned (3)

Actions | Create item

Scan started on March 08, 2026, 21:36:27

1

phone_number *(String)*	seen	number_of_reports	risk_lev
9876543219	298573...	2	HIGH
9876543210	297058...	2	HIGH
9023134211	298586...	1	MEDIUM

Prototype Performance report/Benchmarking

The prototype was tested using simulated call transcripts through the REST API. The system processes requests using AWS API Gateway, AWS Lambda, and DynamoDB with an average response time of **200–500 ms** and total processing time under **1 second**. Fraud indicators such as OTP requests, bank impersonation, and urgent language were correctly detected during testing. The serverless architecture ensures scalability and low latency for real-time fraud analysis.

Future Development

Future improvements include integrating advanced AI/NLP models for more accurate fraud detection, enabling real-time call monitoring, and supporting multiple Indian languages. The system can also be expanded to detect fraud networks using graph-based analysis and integrate with telecom or mobile applications for real-time scam alerts to users.

Estimated implementation cost

Covered under AWS hackathon credits and free-tier usage during prototype development.

Prototype Assets:

- ➔ GitHub Public Repository Link: mightyrudra1511/AIFORBHARAT
- ➔ Demo Video Link: Fraud-Detection-Prototype-Video.mp4 - Google Drive

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Thank You